

Chapter 27 — Inflammatory and Pain Conditions of the Lower Genitourinary Tract: Prostatitis and Related Pain Conditions, Orchitis, and Epididymitis

PROSTATITIS

Historical Perspective

- Knowledge of the prostate traces to **Herophilus**; **Nicola Massa**, a sixteenth-century Venetian physician, is credited with discovering and describing the prostate.
- **First description of prostatitis: Verdies, 1838. Treatment by prostatic massage: Posner of Berlin, 1893** — before which prostatic disease was treated by counter-irritation to the perineum, heat, cupping and leeches.
- **Farman's 1929 American Urological Association (AUA) classification** divided prostatitis by bacterial cause into a **specific (gonorrhoeal)** type and a **non-specific or secondary focal** type, the latter attributed to **haematogenous seeding from other infections**.
- The gonococcal group was subdivided into **simple catarrhal or follicular** (acute infection spreading from the posterior urethra), **true chronic parenchymatous** (from repeated infection), and **atrophic or atonic** — the last held to include **premature ejaculation and impotence**, with causation placed in **neuro- and psychopathology**.
- **Antibiotics materially altered the clinical course of infectious prostatitis.**
- The contemporary history begins with the **Drach letter to the Journal of Urology, 1978** — the first scientific proposal of a systematic classification, built on the **segmented quantitative urogenital cultures of Meares and Stamey, 1968**.
- **Drach's four categories:** acute bacterial prostatitis (ABP); chronic bacterial prostatitis (CBP), defined by **recurrent bacteriuria with the same organism**; nonbacterial prostatitis — **symptoms with negative cultures but inflammatory cells present**; and **prostatodynia** — **symptoms including prostatic discomfort with neither recognisable infection nor inflammation**.

Current Classification of Prostatitis

- Framed at **consensus conferences in 1995 and 1998**; the **National Institutes of Health (NIH) classification** was published in 1999.
- **NIH categories:** I acute bacterial prostatitis; II chronic bacterial prostatitis; IIIA chronic prostatitis/chronic pelvic pain syndrome (CP/CPPS), inflammatory; IIIB CP/CPPS, non-inflammatory; IV asymptomatic inflammatory prostatitis.
- **Categories I and II are carried over essentially unchanged from Drach**; the older **nonbacterial prostatitis and prostatodynia** were folded together into category III. Bacterial prostatitis accounts for 5–10% of cases, leaving 90–95% as category III.
- **IIIA versus IIIB turns solely on leucocytes in expressed prostatic secretions (EPS), post-prostatic-massage urine (VB3) or seminal plasma** — present in IIIA, absent in IIIB. **No clinically significant difference between the two subclasses has been demonstrated**, so the split is classificatory rather than therapeutic.
- **"Prostatitis" is a misnomer in category III:** historically the pain was attributed to prostatic infection or inflammation, but the **prostate may not be the source of the pain at all**, which is why **"chronic pelvic pain syndrome"** was adopted.

- **Category IV is defined by the absence of any genitourinary pain complaint** and is therefore **diagnosed incidentally** — most often on **prostate biopsy performed for suspected cancer** or on **semen analysis performed for infertility**.
- **NIH/Chronic Prostatitis Collaborative Research Network (CPCRN) definition of category III: pain or discomfort in the pelvis for at least 3 of the previous 6 months.**
- **Exclusion criteria:** uropathogenic bacteria on standard microbiological methods, urogenital cancer, prior radiation or chemotherapy, urethral stricture, and neurological disease affecting the bladder.
- The **Multidisciplinary Approach to the Study of Chronic Pelvic Pain (MAPP) Research Network** applied **similar inclusion and exclusion criteria** and reports its work under the term **urologic chronic pelvic pain syndrome (UCPPS)**.

Histopathology

Histology

- **"Prostatitis" carries two distinct meanings** — **inflammation seen on histology** and the **clinical syndromes of genitourinary pain** in the NIH classification. **The relationship between the two is not clear**, and the overlapping usage is itself a source of confusion.
- **Category IV rests on prostate-specific specimens** — EPS, VB3, seminal plasma and biopsy tissue — and **"asymptomatic" here means specifically the absence of pain.**
- **Histological inflammation is a common incidental finding**, in tissue resected for **benign prostatic hyperplasia (BPH)** and in biopsies taken for suspected cancer. **Autopsy series place chronic inflammation in over 70% of prostates** — close to a **background state of the adult prostate**, and therefore **unable by itself to define a clinical entity**.
- **Intraepithelial lymphocytes are increased in areas of category IV prostatitis relative to normal prostate**, raising an **immune or autoimmune origin** rather than an infective one.

Nickel classification of prostatic inflammation.

- Reports inflammation by **three parameters** — **anatomical location, extent and grade** — rather than as a single label.
- **Location: glandular** — within the duct or gland epithelium or lumen; **periglandular** — within stroma but centred on ducts and glands, within approximately **50 µm** of a gland; **stromal** — in the stroma and **more than 50 µm** from a gland.
- **Extent: focal <10%, multifocal 10–50%, diffuse >50%.**
- **Grade: 1 (mild)** — individual inflammatory cells separated by distinct spaces; **2 (moderate)** — confluent sheets with no tissue destruction and no lymphoid follicles; **3 (severe)** — confluent sheets with tissue destruction or nodule formation. **The qualitative discriminator, not the cell count, separates the grades:** absence of confluence → confluence without destruction → confluence with destruction or nodularity.

Associated entities.

- **Corpora amylacea** form from **deposition of prostatic secretions around sloughed epithelial cells**. They are **not inflammatory in themselves** — they cause inflammation only when they obstruct.
- **Prostatic calculi** contribute in two ways: **local obstruction** and **acting as a nidus for bacterial growth**. The **chemical composition of many such stones points to chronic infection** as their origin.
- **Stones correlate with the symptoms of CP/CPPS**, and **occasionally grow large enough to obstruct the urethra** and require surgical removal.

Specific Cases of Prostatic Inflammation: Granulomatous Prostatitis

- **Diagnosed histologically by epithelioid granulomas, with or without other inflammatory cells** — a tissue diagnosis, not a clinical one.
- **Encountered incidentally on transurethral resection specimens and prostate biopsies.**
- **Epstein and Hutchins classification** — four categories: specific, non-specific, post-transurethral resection of the prostate (TURP), and allergic granulomatous prostatitis.
- **Most characteristically follows intravesical bacillus Calmette-Guérin (BCG) therapy for bladder cancer**, where it produces a **transient rise in prostate-specific antigen (PSA)** — a benign, self-limiting cause of PSA elevation that must not be read as cancer.
- **Tuberculosis (TB) is also a cause.**

Category I: Acute Bacterial Prostatitis

Epidemiology and Routes of Infection

- **Bimodal age distribution:** a peak at 20–40 years and a second peak after 60.
- **Ascending urethral infection** is the principal route — supported by finding **the same organism in the patient's urine and in the rectal swab of the patient or sexual partner.**
- **Other routes:** direct seeding at prostate biopsy; intraprostatic reflux of infected urine into the ejaculatory and prostatic ducts of the posterior prostatic urethra; rarely, lymphatic spread from the rectum; and haematogenous dissemination in bacterial sepsis or other blood-borne infection such as TB.

Risk factors.

- **Sexual:** unprotected intercourse, specifically **insertive anal intercourse.**
- **Anatomical and appliance-related:** phimosis, condom catheter use, indwelling urethral catheters.
- **Instrumentation:** endoscopic procedures and prostate biopsy.
- **Obstruction and stasis:** dysfunctional voiding, distal urethral stricture, BPH.
- **Antecedent infection:** bacterial cystitis or epididymo-orchitis.
- **Clean intermittent catheterisation** — **up to one-third develop prostatitis.**

Presentation

- **Acute lower urinary tract infection (UTI) symptoms** — frequency and dysuria — **are the core, with pain in the lower abdomen and perineum.**
- **Systemic features** — malaise, fever, myalgia.
- **Urinary retention may occur by three mechanisms:** **acute swelling of the prostate, pain, or bladder neck spasm.**
- **Severe cases present as sepsis:** high fever and rigors, cardiovascular instability, altered mental status.
- **Spontaneous versus instrumentation-related disease:** spontaneous ABP occurs in **younger patients with lower septicaemia risk**; ABP following lower urinary tract manipulation or biopsy occurs in **older patients with a greater likelihood of septicaemia**, increased further where the organism is **extended-spectrum β -lactamase (ESBL)-producing.**
- **ABP must be considered alongside pyelonephritis and acute cystitis in any febrile UTI.** In febrile UTI, **over 90% show a rise in PSA and/or prostate volume** even when the picture — including flank pain — suggests pyelonephritis: **prostatic involvement is the rule rather than the exception in febrile male UTI.**

Microbiology

- ***Escherichia coli* is the commonest causative organism, 65–80% of cases.**

- Other Gram-negatives: *Pseudomonas aeruginosa*, *Proteus mirabilis*, *Klebsiella* and *Serratia* species.
- *Enterococcus* species account for up to 10%.
- *Neisseria gonorrhoeae* must be considered in the sexually active young patient.
- **Rare causes:** obligate anaerobes and Gram-positive organisms other than *Enterococcus*.
- *Mycobacterium tuberculosis* is a rare cause, usually with immunodeficiency. The prostate is involved in approximately 10% of genitourinary TB, presenting with perineal pain and dysuria; tuberculous prostatitis can cavitate into abscesses communicating with perineal sinuses and producing fistulae.
- In impaired immunity, notably human immunodeficiency virus (HIV) infection, the list widens to *Staphylococcus aureus*, *Klebsiella pneumoniae*, *P. aeruginosa*, *Serratia marcescens*, *Salmonella typhi*, *M. tuberculosis* and *Mycobacterium avium intracellulare*.

Virulence factors.

- **ABP in previously healthy young men is driven by bacterial virulence rather than host susceptibility** — these are men in whom UTI is otherwise uncommon, so the organism, not the host, explains the infection.
- Uropathogenic *E. coli* from ABP carries a large complement of virulence genes spanning **adhesins, toxins and siderophores**, with expression of **α -haemolysin, aerobactin, cytotoxic necrotising factor-1, cytolethal distending toxin type 1 and colicins**.
- **Biofilm production is the second key mechanism.** A biofilm is an aggregate of bacteria protected by a polysaccharide matrix, and its significance is that **it impedes antibiotic penetration. Biofilm formation is commoner in *E. coli* from acute prostatitis than from cystitis or pyelonephritis** — a structural reason why prostatic infection needs longer, better-penetrating therapy.

Cause determines microbiology and susceptibility.

- **Community-acquired versus nosocomial:** community-acquired ABP is roughly **three times commoner**. *E. coli* predominates in both, but **nosocomial cases are more often *P. aeruginosa*, enterococci or *S. aureus*, with greater resistance and more clinical failures**.
- **Spontaneous versus post-instrumentation:** *E. coli* again predominates, but **post-manipulation disease has a much higher prevalence of *Pseudomonas* (20%) and a higher risk of prostatic abscess**.
- **After transrectal biopsy:** organisms are **more resistant to fluoroquinolones, more likely ESBL-producing, and more likely to yield positive blood cultures**.
- **Antibiotic selection in ABP must therefore be dictated by the route of infection, not by a single empirical default.**

Evaluation

- **The first historical determination is whether the prostatitis is spontaneous or follows lower urinary tract manipulation or recent transrectal prostate biopsy** — this single point drives the empirical antibiotic choice.
- **Complicating factors to seek:** diabetes mellitus, HIV infection, neurological disease, recent antibiotic use.
- **A palpable bladder suggests retention. ABP is the one situation in which a genuinely "boggy" prostate may be palpated, from oedema; the prostate is tender and swollen in 60–90% of cases.**
- **Aggressive palpation must be avoided** — it risks bacterial dissemination and sepsis. **Prostatic massage for expressed secretions is contraindicated acutely for the same reason.**
- **Full blood count, urinalysis and midstream urine culture; urine culture positive in 60–85%;** urinalysis shows white and red cells.
- **Blood cultures before antibiotics in any patient with fever or rigors.** Positive cultures are associated with **fever above 38.4 °C (101.1 °F)**, usually grow the same organism as the urine, and **add to the microbiological diagnosis only in a small minority** — their value is in the septic patient, not routinely.

- **Nucleic acid amplification testing for gonorrhoea and chlamydia if urethral discharge is present; sexually transmitted infection prevalence in ABP is only about 2%.**
- **Renal function to guide dosing.**
- **Post-void residual assessment to exclude retention — preferably non-invasively by ultrasound, since catheterisation of an acutely infected prostate is itself hazardous.**
- **PSA is not recommended acutely because the level is expected to be raised. If measured and elevated, follow it to normalisation — which may take up to 6 months — to exclude underlying cancer.**
- **Imaging is not indicated unless abscess is suspected; then transrectal ultrasonography (TRUS) or computed tomography (CT). There is no role for prostate biopsy in acute prostatitis.**

Treatment

- **ABP must be treated fully, not only to resolve the acute illness but to prevent chronic infection — approximately one in ten progresses to category II and a further one in ten to category III. ESBL infection after transrectal biopsy is itself a risk factor, with about a quarter progressing.**
- **Admit for parenteral antibiotics, hydration and laboratory monitoring where there are systemic signs. Outpatient treatment only if all three hold: no systemic illness, tolerating oral intake, no urinary retention.**
- **European Association of Urology (EAU) guidance for complicated UTI with systemic symptoms — parenteral high-dose bactericidal therapy: amoxicillin plus an aminoglycoside; a second-generation cephalosporin plus an aminoglycoside; or a third-generation cephalosporin.**
- **Ciprofloxacin is restricted to situations where local resistance is below 10%, the entire course can be given orally, the patient does not require hospitalisation, or the patient has had anaphylaxis to β -lactams.**
- **Post-transrectal-biopsy ABP is the specific exception: given fluoroquinolone resistance and ESBL rates, a carbapenem is strongly argued for in any patient with fever and prostatitis after transrectal biopsy.**
- **AUA White Paper on complications after prostate biopsy — for the febrile post-biopsy patient: do not offer fluoroquinolones or trimethoprim-sulfamethoxazole (TMP-SMX); manage with aggressive resuscitation and broad-spectrum cover — carbapenems, amikacin, or second- and third-generation cephalosporins — after urine and blood cultures.**
- **Oral fluoroquinolone where the organism is susceptible, either once fever has subsided or as initial therapy in those safe for discharge: ciprofloxacin 500 mg twice daily or levofloxacin 500 mg once daily.**
- **Reculture the urine after one week to confirm clearance.**
- **Duration: the traditional recommendation was 4 weeks, but 2 weeks of ciprofloxacin gives comparable bacteriological cure and 1-year clinical cure, so 2 weeks of oral therapy is likely sufficient.**
- **Relapse is lower after fluoroquinolones and parenteral β -lactams than after co-trimoxazole and oral β -lactams.**
- **Tuberculous prostatitis: anti-tuberculous chemotherapy for at least 6 months.**

Adjuncts and bladder drainage.

- **Non-steroidal anti-inflammatory drugs (NSAIDs) for pain and inflammation; α -adrenergic blockers where lower urinary tract symptoms (LUTS) are present.**
- **Where retention requires drainage, the traditional position is a suprapubic catheter, to avoid irritating the prostate and provoking bacteraemia. Short-term drainage may be by intermittent or brief urethral catheterisation, though not all authors accept the urethral route even short-term. A suprapubic catheter is recommended for long-term drainage — this is the settled point.**

Prostatic Abscess

- **Suspect on high fever, immunosuppression, or — the practical trigger — failure to respond to initial therapy after 48 hours.**
- **Risk factors:** prior or indwelling catheter; previous genitourinary surgery, lower urinary tract instrumentation, prostate biopsy; bladder outlet obstruction; acute and chronic bacterial prostatitis; increasing age and comorbidity, diabetes, cirrhosis, HIV, chronic renal failure and haemodialysis.
- **Imaging: TRUS shows a hypoechoic area; CT gives clarity of location, preoperative planning and identification of extraprostatic spread** — the deciding consideration where drainage is contemplated.
- **Small abscesses (<1–2 cm) may be treated conservatively with antibiotics;** note that **surgical drainage of a small abscess may permit a shorter antibiotic course than conservative management.**
- **Progression of symptoms mandates intervention.**
- **Localised or very peripheral lesions → percutaneous drainage under ultrasound guidance; recurrence after aspiration is 15–33%.**
- **Failure of percutaneous drainage, or lesions too large to drain percutaneously → TURP to unroof the abscess.** Intraoperative ultrasound may localise the cavity, and **palpation and massage of the prostate after unroofing helps complete drainage.**
- **Extraprostatic extension → open surgical drainage (rare).**

KEY POINTS: ACUTE BACTERIAL PROSTATITIS

- Category I presents with acute dysuria, frequency and possibly fever.
- The commonest causative organism is *E. coli*.
- Organisms and sensitivities differ by cause — spontaneous, after transurethral procedure, or after transrectal biopsy.
- ABP after transrectal biopsy should be treated with carbapenems, amikacin, or second- and third-generation cephalosporins.
- Use a suprapubic tube if long-term drainage is needed.
- Failure to respond to therapy should prompt imaging for a prostatic abscess.

Category II: Chronic Bacterial Prostatitis

Definition and Course

- **Defined by recurrent UTIs with the same organism, implying a persistent prostatic source** — the organism is not being reacquired, it is being harboured.
- **Accounts for 5–10% of prostatitis cases.**
- **Dysuria and pain generally respond to antibiotics,** and — the discriminating feature against category III — **patients are relatively asymptomatic between episodes,** whereas CP/CPPS is continuous.

Bacteria Causing Category II Prostatitis

- **Established:** *E. coli*, *Pseudomonas*, *Proteus*, *Klebsiella* and *Enterobacter* species.
- **Enterococcus is generally accepted as a cause,** with *Enterococcus faecalis* the dominant Gram-positive isolate in large prostatitis-clinic series.
- **Debatable significance:** staphylococci and streptococci, *Corynebacterium*, *Ureaplasma urealyticum* and *Mycoplasma hominis*.
- **Fastidious organisms requiring different culture techniques:** *M. tuberculosis* and *Candida* species — they will be missed on routine culture.

- ***Trichomonas vaginalis***, a flagellate protozoan, is included among possible causes, since trichomoniasis can present as urethritis, epididymitis or prostatitis.
- **Biofilm formation occurs as it does in category I** and is the reason for poor treatment response — the same mechanistic link as in acute disease.

Role of Chlamydia in Prostatitis

- ***Chlamydia* is an obligate intracellular Gram-negative organism requiring living host cells to multiply.**
- **Ascending prostatic infection after urethral inoculation has been demonstrated in animal models.**
- **The central objection is site attribution, not detection:** *Chlamydia* is a common cause of urethritis, so its presence in EPS or semen **cannot be attributed to the prostate rather than to urethral contamination.**
- The same objection applies to tissue: **TURP chips, transrectal biopsies and open surgical specimens are all open to contamination from the prostatic urethra.**
- **Contamination is eliminated only by perineal biopsy of the lateral lobes** — the methodological crux of the debate — and **the two studies using lateral-lobe biopsies detected no *Chlamydia*.**
- **Counter-evidence:** ejaculate plasmid DNA and **antichlamydial mucosal immunoglobulin A (IgA)** found in patients and not controls, with **IgA levels correlating with symptoms and interleukin-8** — a demonstrable local immune response rather than mere presence of organism.
- ***Chlamydia*-attributed chronic prostatitis carries a roughly threefold higher prevalence of premature ejaculation (PE)** than prostatitis from common uropathogens.
- **The most compelling evidence is the treatment-response study:** patients with *Chlamydia* in EPS or VB3 but a **negative first-voided urine (VB1)** — localised to the prostate, not the urethra — treated with 4 weeks of **levofloxacin plus azithromycin** showed **eradication with a fall in serum PSA and reduction in symptoms.** A PSA response implies the prostate was the infected compartment.
- **Conclusion:** despite continuing objections, some evidence supports a causative role for *Chlamydia* in CBP.

Diagnosis and Evaluation

- **Current standard: the pre-massage and post-massage (two-glass) test.** Midstream pre-massage urine, then **the initial 10 mL of urine voided after prostatic massage**; both cultured. **The VB3 specimen demonstrates a persistent prostatic source.**
- **Previous standard: the Meares-Stamey four-glass test** — **VB1** (first voided urine, urethral bacteria), **VB2** (midstream urine, bladder), **EPS** (expressed prostatic fluid), **VB3** (post-massage urine).
- **The two-glass test is simpler and yields similar results** — it predicts the four-glass result in over 95% of patients, and is the practical replacement.
- **No diagnostic cut-off for bacterial counts is recommended**, though some clinics use a tenfold rise in the post-massage specimen.
- **Semen cultures are not recommended** — poor detection of causative bacteria and reduced specificity from skin contamination.
- **In HIV infection, send cultures additionally for anaerobes, aerobes, fungi and TB.**
- **Haematuria with infection must be rechecked 4–6 weeks after resolution; persistent haematuria requires full evaluation.**
- **Examination:** fever; abdominal examination to exclude other causes of suprapubic pain; scrotal examination for epididymal or testicular inflammation; **digital rectal examination (DRE) for prostate size and for nodularity suggesting cancer.**

- **Bladder outlet obstruction (BOO) and urinary retention must be sought**, since either predisposes to recurrent UTI. A post-void residual above 180 mL correlates with increased infection risk, and retention additionally increases bacterial pathogenicity.
- **Not all patients need imaging.** Men under 45 with a first episode of acute UTI show no abnormalities on imaging, but **they do require assessment for urethral stricture**, which complicates **up to 41% of UTIs**.
- **Image where any of the following is present:** UTI with diabetes mellitus, chronic kidney disease, stones, voiding difficulty, neurological disease, poor response to antibiotics, infection with urea-splitting organisms, or haematuria persisting beyond 1 month.

Treatment of Category II Chronic Bacterial Prostatitis

- The prostatic capillary bed has no active transport mechanism for antibiotics, so entry is entirely by passive diffusion.
- Passive transport is governed by drug concentration, lipid solubility, degree of ionisation or charge, protein binding, and molecular size and shape.
- Quinolones penetrate excellently because they are zwitterions — carrying both positively and negatively charged groups — and are highly lipid-soluble. The fluid in CBP is alkaline, which further concentrates quinolones.
- Other agents with good to excellent penetration: tetracyclines (lipid solubility), macrolides (favourable pKa), and trimethoprim (a lipid-soluble base with favourable pKa).
- There are no AUA guidelines for antibiotic treatment of CBP; the EAU guideline is the reference:
 - Fluoroquinolone for 4–6 weeks, first line — no difference between ciprofloxacin and levofloxacin.
 - *Trichomonas vaginalis*: metronidazole 500 mg three times daily for 14 days.
 - Obligate intracellular pathogens (*Chlamydia trachomatis*): azithromycin 500 mg daily for 3 weeks.
 - Doxycycline 100 mg twice daily for 10 days.
- Fluoroquinolones are chosen for two reasons, not one: tissue penetration and spectrum — Gram-negatives including *Pseudomonas*, Gram-positives, and atypicals including *Chlamydia* and genital *Mycoplasma*.
- The 4–6 week duration rests on expert opinion and agrees with the UK Prostatitis Expert Reference Group guidance.
- Adding a macrolide may help because macrolides penetrate biofilms — the therapeutic counter to the biofilm mechanism. A higher-dose intermittent azithromycin regimen with ciprofloxacin over 4 weeks improved eradication over the standard daily schedule given for 6 weeks.
- Quinolone regimens cure 60–80% of *E. coli* and other Gram-negative Enterobacteriaceae, against only 30–50% for a 12-week course of TMP-SMX.
- Fluoroquinolone-resistant but TMP-SMX-susceptible → 3-month course of TMP-SMX. There are no evidence-based recommendations where the organism is resistant to both.
- Cochrane conclusions: no difference among the oral fluoroquinolones; no conclusion possible on optimal duration within 4–12 weeks; and for chlamydial prostatitis, azithromycin is superior to ciprofloxacin and equivalent to clarithromycin.

Chronic Bacterial Prostatitis in HIV and Immunocompromised Patients

- Consider low-dose antimicrobial suppression for a period to reduce recurrence.
- Where the patient remains persistently immunocompromised despite antiretroviral therapy, lifelong suppressive antimicrobials have been recommended to minimise progression to prostatic abscess.

KEY POINTS: CHRONIC BACTERIAL PROSTATITIS (CATEGORY II)

- CBP is characterised by recurrent UTIs with the same organism.
- Unlike category III CP/CPPS, patients are relatively asymptomatic between episodes.
- Urinary retention as a cause of recurrent UTI must be excluded.
- Patients under 45 do not need imaging but need assessment for a urethral stricture.
- Treatment is with antibiotics that achieve adequate prostatic concentrations.
- Daily suppressive antibiotics or TURP can be used for refractory cases.

CATEGORY III: CHRONIC PROSTATITIS/CHRONIC PELVIC PAIN SYNDROME**Aetiology**

- The cause and much of the pathogenesis remain unknown despite 25 years of concerted research.
- Evidence exists for five contributing domains: infection, neurological, inflammatory/autoimmune, endocrine and psychological. It is the interplay between them that creates and sustains pelvic pain.
- The unifying hypothesis: an insult — infection, stress or trauma — in a genetically susceptible individual initiates neurogenic inflammation, which is then maintained by the other factors. Initiation and response are separable, and the maintaining mechanism need not be the initiating one.

Infection

- The assumption of infection is driven by symptom similarity to true prostatic infection — it is not supported by microbiological evidence.
- Prior sexually transmitted infection (STI) raises the odds of prostatitis roughly 1.8-fold, yet there is no evidence of active STI — a historical association, not ongoing infection.
- Routine cultures of urine, EPS, post-massage urine and seminal plasma do not differ between patients and asymptomatic controls; around 70% of both groups harbour some bacteria, establishing that bacteria are routinely present in the prostate of asymptomatic men and do not by themselves produce disease.
- Polymerase chain reaction (PCR) has failed to identify any clear causative organism.
- Broad-range bacterial DNA amplification with sequence matching — detecting species above about 1–3% of the biome — found only one significant difference: *Burkholderia cenocepacia* over-represented in VB1 specimens. This is an opportunistic pathogen of cystic fibrosis known to form biofilms; note that the finding was in the urethral specimen, not a prostatic one.
- Animal model evidence is the conceptually important item: the CP1 strain, isolated from a patient, induced and sustained chronic pelvic pain that persisted after the bacteria had been cleared from the genitourinary tract. Bacteria may therefore initiate a pain state that becomes independent of their presence — which explains both the infective association and the failure of antimicrobial therapy.
- Microbiome findings run in opposite directions in the two compartments: increased bacterial diversity in the urinary tract, reduced diversity in the gut.

Inflammation

- Only about one-third of patients with clinical CP/CPPS have inflammation on prostate biopsy.
- Where inflammatory cells are present in VB3, EPS or seminal plasma, the degree of inflammation does not correlate with symptoms — reinforcing that the IIIA/IIIB distinction is not clinically meaningful.

- **REDUCE:** inflammation on initial prostate biopsy **did not predict the incidence of CP/CPPS symptoms**; however, among those with the condition, **chronic — but not acute — inflammation was associated with a shorter time to symptom progression**. Chronic inflammation is prognostic, not diagnostic.
- Individual cytokine data vary widely and are not conclusive.
- Autoimmunity is supported by an increased lymphoproliferative response to prostatic antigens; the identified targets are a region of prostatic acid phosphatase, PSA, and human seminal vesicle secretory protein 2.
- Chemokines: monocyte chemoattractant protein-1 and macrophage inflammatory protein-1 α (MIP-1 α) are elevated relative to men with no urological disease or with BPH, and MIP-1 α correlates with both total NIH Chronic Prostatitis Symptom Index (NIH-CPSI) score and its pain component.
- Interleukin-17 is elevated and correlates with symptoms.
- The examinable pattern: the molecules that correlate with symptoms are chemokines and cytokines, not histological inflammation.

Neurologic Causes

- Pain is the cardinal symptom and is mediated through the nervous system.
- Self-reported neurological disease is commoner than in controls, and autonomic nerve function differs.
- There is evidence of central sensitisation in both afferent and efferent pathways — defined as continued nerve activity in the absence of stimulation, or activity sustained by only a very low level of nociceptive stimulation.
- Sensory perception thresholds are not different, indicating no peripheral neuropathy — the lesion is central, not peripheral.
- Functional magnetic resonance imaging of spontaneous pelvic pain — patients scanned for 10 minutes rating their own pain with no external stimulus — localised pain perception to the anterior insula, correlating with the intensity of self-reported pain.
- Grey-to-white matter relationships differ from controls, and these findings differ from those in other pain conditions such as low back pain — the central signature is condition-specific, not generic to chronic pain.
- Lower white matter tract density in regions subserving perception, integration of sensory information and pain modulation.
- Alterations occur in the default mode network — based on the medial prefrontal cortex and posterior cingulate gyrus, more active at rest than during tasks — which may explain pain interference during resting moments.
- Patients with widespread pain show increased grey matter volume and functional connectivity involving the sensorimotor and insular cortices, identical to fibromyalgia — centralised pain in this subgroup is the same phenomenon as fibromyalgia.

Pelvic Floor Dysfunction

- The postulated link is neurological: nerve abnormality producing pelvic floor spasm.
- Two consistent findings: pathological tenderness of the striated pelvic floor muscle, and poor or absent ability to relax the pelvic floor on single or repetitive effort — it is the failure of relaxation, not weakness, that characterises the condition.
- Electromyography shows greater resting hypertonicity and instability with lowered voluntary endurance contraction amplitude — high at rest, poor on demand. Transabdominal ultrasound shows reduced mobility due to increased tension.

- **Prevalence of tenderness depends on how it is sought.** Early CPCR examination found internal and external tenderness in only a small minority; the **MAPP standardised examination by trained examiners palpating six defined sites found tenderness in 81% of participants against 9% of controls**, with about one in five tender at all six sites. **Tenderness is missed when not systematically sought.**
- **Tenderness burden tracks disease burden:** greater tenderness correlates with worse pelvic pain, worse urinary symptoms, worse quality of life (QOL) and more widespread non-pelvic pain.
- **Neural correlates: decreased resting-state connectivity between the motor areas controlling the pelvic floor and the right posterior insula**, a region handling **pain processing and sympathetic autonomic control**; **less connectivity is associated with greater pain**. These connections can be altered by a **history of adverse early life events**.
- **The pelvic floor is activated in synergy with the gluteal muscles** as well as by its direct cortical innervation — therefore **gluteal and gait abnormalities may drive abnormal pelvic floor activation**, the rationale for assessing beyond the pelvis.

Psychosocial Factors

- **Greater perceived stress is associated with greater pain intensity and disability.**
- **Helplessness and catastrophising predict overall pain, urinary symptoms and depression** — cognitive style is predictive, not merely reactive.
- **Systematic review confirms pain intensity is associated with catastrophising, perceived stress, and low satisfaction with relationships including sexual functioning.**
- **A history of sexual, emotional or physical abuse is associated with increased risk of CP/CPPS symptoms (odds ratios 1.7–3.3).**

Endocrine Abnormalities

- **Alteration of the hypothalamic–pituitary–adrenal axis is common to chronic pain conditions and is reported here.**
- **The cortisol awakening rise is exaggerated relative to controls.**
- **Baseline adrenocorticotrophic hormone (ACTH) is lower, and the ACTH rise in response to stress is blunted.**
- **The pattern is dissociated — exaggerated cortisol response on waking with a blunted ACTH stress response** — not a uniform up- or down-regulation of the axis.

KEY POINTS: AETIOLOGY OF CP/CPPS

- Patients with CP/CPPS have the same quantity of bacteria in urine, prostatic fluid and seminal plasma as those without pain.
- There is evidence of central sensitisation of the central nervous system and altered pain processing.
- Where inflammation is present in VB3, EPS or seminal plasma, its degree does not correlate with symptoms.
- Helplessness and catastrophising predict overall pain, urinary symptoms and depression.
- Baseline ACTH is lower and the ACTH rise to stress blunted compared with asymptomatic individuals.
- Patients with CP/CPPS have increased pain sensitivity compared with controls.

Symptoms in Chronic Prostatitis/Chronic Pelvic Pain Syndrome

- **Pain is the symptom that distinguishes category III from other conditions such as BPH** — the discriminating feature, not the voiding symptoms.

- **Early symptom cataloguing** established the hierarchy still quoted: **pelvic pain both the most frequent and the most severe symptom**, followed by urinary frequency and obstructive voiding symptoms, with the **perineum the commonest site of pain**.

The NIH Chronic Prostatitis Symptom Index.

- **Developed by the NIH as an outcome measure for clinical trials**, derived from focus groups of patients with pelvic pain in the absence of infection; also useful for categorising symptoms.
- **Three domains: pain, urinary symptoms and quality of life.**
- **The pain domain covers six items: perineum, lower abdomen/suprapubic area, testes, penis, pain with ejaculation, and dysuria.**
- **Validated, with the pain and QOL subscales sensitive to change** — which is why they, rather than the urinary subscale, serve as trial endpoints.

Prevalence of Individual Symptoms

- **Mean NIH-CPSI score in large cohorts: 20.4.**
- **Pain sites in descending frequency: perineum 63.3%, testicles 57.6%, suprapubic area 42.3%, tip of penis 32.4%.**
- **Pain with ejaculation 47.7% and dysuria 43.3%, and the two are associated.**
- **A greater number of pain locations predicts greater pain severity, worse voiding symptoms and worse QOL** — pain site count functions as a severity marker.
- **Frequency and intensity of pain have the greatest impact on QOL, whereas urinary symptoms do not correlate with QOL** — the first signal that pain and urinary symptoms are separable constructs.
- **Pain severity distribution: mild 32%, moderate 52%, severe 16%.**

Bladder Pain in CP/CPPS

- **The long-held tenet that interstitial cystitis (IC)/bladder pain syndrome (BPS) is overwhelmingly female** — an oft-quoted 10:1 female-to-male ratio — is challenged by the MAPP data.
- **Three-quarters of males in the first MAPP study had either painful urgency or pain worsened by bladder filling. Painful urgency is specifically urgency due to pain, pressure or discomfort, not fear of leaking** — the definitional distinction from overactive bladder.
- **Bladder-focused symptoms mark a more severe phenotype:** more flares, higher catastrophising, more frequently reported irritable bowel syndrome (IBS), and more severe pain, frequency and urgency.
- **17% of males meeting the high-specificity IC/BPS definition also met the CP/CPPS case definition** — considerable overlap between the two conditions as currently defined.
- **Hunner's lesion: an area of inflammation that may respond to local ablative therapy** — the one phenotype with a directed treatment. **No difference in incidence between males and females; overall 12%, rising with age, particularly over 50.** Those with lesions were more likely to report painful urgency.

Pain Beyond the Pelvis

- **Pain extending beyond the pelvis marks more severe disease than pelvic pain alone.**
- **MAPP used a whole-body map of 45 sites in 7 zones.**
- **Non-pelvic pain severity increases with the number of regions reported**, and a greater number of sites is associated with **greater sleep disturbance, depression and anxiety, psychological stress, somatic symptom burden and negative affect**, and **more symptoms of IBS, fibromyalgia and migraine.**

Chronic Overlapping Pain Conditions

- **Individuals with chronic overlapping pain conditions (COPCs) have more severe symptoms than those with urological symptoms alone.**

- **Approximately one-third of MAPP participants with pelvic pain had one or more COPCs, the commonest being IBS, with more severe urological symptoms and more frequent depression and anxiety.**

Relationship Between Pain and Urinary Symptoms

- **The traditional tenet was that pain and urinary symptoms share a cause and can be measured together. MAPP overturned this.**
- **Principal components and exploratory factor analysis showed two separate factors — pelvic pain and urinary symptoms.**
- **Depression was predicted by pain but not by urinary symptoms, confirming the two domains differ in their impact.**
- **Pain and urinary symptoms must therefore be assessed separately rather than combined, as they are in the NIH-CPSI and the Genitourinary Pain Index (GUPI).**
- **MAPP drew items from the GUPI and the IC Symptom Index to construct separate pain and urinary assessments, which have been shown valid and useful in quantifying outcomes and predicting symptom change; the two show significant but independent associations with non-pelvic pain severity, fatigue and depression.**

Fluctuations in Symptom Severity

- **Tracked every 2 weeks for 1 year, improvement was commoner than worsening in both domains, and pain fluctuated more than urinary symptoms in both directions.**
- **Predictors of worsening overall symptoms at 1 year: extent of widespread pain, presence of COPCs, and poorer overall health.**
- **Predictors of worsening pain specifically: pain catastrophising and self-reported stress.**
- **Anxiety, depression and general mental health did not contribute to symptom worsening — counterintuitive given how strongly they associate cross-sectionally.**
- **Follow-up over a further 2–9 years showed minimal additional change — the condition is largely stable once established.**
- **The distinction between patients with and without pain outside the pelvis is a clinically meaningful phenotyping tool, and the two groups remain stable over 12 months — a trait, not a fluctuating state.**
- **Flares are significant symptom exacerbations; roughly 70% flare at least once in a year. Only recent sexual activity and UTI-like symptoms predicted flares; diet, physical activity, sedentary behaviour, stress and constipation did not.**

Sexual Dysfunction

- **Erectile dysfunction (ED) prevalence is 15–40%, higher with more severe symptoms; moderate to severe ED affects around 29% of those with moderate to severe symptoms, and only about one in ten reports no ED at all.**
- **The association runs both ways: men with an ED diagnosis are significantly more likely to have been previously diagnosed with CP/CPPS.**
- **Ejaculatory pain is reported at some point by about three-quarters, with premature ejaculation at similar levels.**
- **Sexual dysfunction is worse than in IBS, fibromyalgia or chronic fatigue used as positive controls — lower International Index of Erectile Function–Erectile Function domain scores, more ejaculatory dysfunction and poorer quality of sexual relationships. The sexual morbidity is specific to this condition, not generic to chronic pain.**

Anxiety and Depression

- **Approximately twice as likely to report anxiety and depression as age-matched controls.**
- **Greater current and lifetime stress, poor illness coping and increased self-reported cognitive deficits** compared with age- and education-matched controls.
- **Odds of an anxiety disorder diagnosis roughly doubled (odds ratio 2.10), across all age groups and strongest at 40–59 years (odds ratio 2.53).**
- **Mood may mediate the relationship between pain and urinary symptoms:** pelvic pain and urinary symptoms are **bidirectionally** related to each other and to negative mood and reduced QOL.

Phenotypic Approach to Symptoms: UPOINT

- **The phenotype is variable and typically involves multiple symptom types** — no single presentation defines it.
- **UPOINT has six clinically defined domains: Urinary, Psychosocial, Organ-specific, Infection, Neurologic/systemic, and Tenderness of skeletal muscle.**
- **The impetus was the failure of monotherapy in CP/CPPS clinical trials** — if patients differ in which domains are active, a single agent tested across all of them will fail. **The purpose is to target therapy to the domains a given patient occupies**, and it also serves as an **aide-mémoire** for the assessment itself.
- **Domain distribution: only about one in five is positive for a single domain, about one-third for two, and the remainder for three or more.**
- **Strong correlation between the number of domains and symptom severity** — domain count is a severity index as well as a treatment map.
- **The "snowflake hypothesis": each patient has a unique phenotype**, the conceptual justification for multimodal, individualised therapy.
- **Both the UK guidelines and the ICUD summary recommend UPOINT or an equivalent** to phenotype symptoms and direct therapy.

Evaluation of Chronic Prostatitis/Chronic Pelvic Pain Syndrome

- **CP/CPPS is a diagnosis of exclusion; the evaluation must rule out identifiable causes of pelvic pain.**
- **To meet the NIH consensus definition the patient must not have active urethritis, urogenital cancer, urinary tract disease, functionally significant urethral stricture, or neurological disease affecting the bladder.** A full list of inclusion, exclusion and deferral criteria used in NIH trials was published by the CPCRNI.
- **Many comorbidities lie outside the usual scope of urology — neurological, immunological and psychological** — the practical argument for multidisciplinary management.
- **Three guideline sources inform evaluation and treatment:** the **UK Prostatitis Expert Reference Group**, the **International Consultation on Urologic Disease (ICUD)**, and the **EAU**.

History

- **The circumstances at onset are the highest-yield element**, because they point to the likely domain:
 - **Onset with significant stress or psychological trauma → more likely pelvic floor dysfunction.**
 - **Onset after physical trauma → neurological damage or musculoskeletal abnormality.**
 - **Onset after UTI or STI → assess for persistence of infection.**

Assessment

- **NIH-CPSI** covers pain location, frequency and severity; voiding symptoms; and interference/QOL. **Its principal limitation is that it contains no questions about bladder-related pain** — serious, given that around 42% of MAPP males met criteria for IC/BPS.
- **The GUPI is a modification adding two questions on pain with bladder filling or emptying**, and therefore captures bladder symptoms better. **Identifying bladder pain prompts bladder-specific treatments, which are of no use in patients without bladder pain.**
- **The separation of pain from urinary symptoms as an outcome measure matters chiefly for trials; for practice the GUPI is a convenient single inventory.**
- **Voiding:** both instruments contain only two voiding questions, so **the AUA Symptom Index is used to assess voiding dysfunction properly.**
- **Sexual function:** cover ED, libido and ejaculatory problems; the Sexual Health Inventory for Men (SHIM), a shortened International Index of Erectile Function, quantifies ED.

Physical Examination

- **Neurological:** screening examination for abnormalities. **Abdominal:** exclude other causes of abdominal or suprapubic pain.
- **Genitourinary:** hernia, hydrocele, testicular masses, penile lesions; DRE for rectal masses and haemorrhoids.
- **On DRE the prostate is tender in fewer than half of patients** — prostatic tenderness is not required for the diagnosis. **Severe tenderness suggests acute prostatitis. Nodularity must never be attributed to inflammation and must prompt consideration of prostate cancer.**
- **Pelvic musculature:** during DRE, **palpate the muscles lateral to the prostate and extending to the coccyx** to identify **myofascial trigger points**, which identify the patients who will benefit from **pelvic floor physical therapy and relaxation techniques**. **Technical manoeuvre:** have the patient abduct the leg on each side to make the muscles easier to palpate.
- **The extended examination covers four sites:** the perineal body (midway between the anus and the inferior edge of the scrotum), the levator (posteriorly), the obturator (anteriorly and laterally), and the urogenital diaphragm (anteriorly at the prostatic apex).
- **The extended examination detected pelvic floor tenderness in 47% of patients against 6% of controls**, while prostatic tenderness was present in only a small number of cases and in none of the controls — the tenderness is muscular, not prostatic.

Laboratory and Office Studies

- **Urinalysis:** a positive dipstick for haematuria must be confirmed by **≥3 red blood cells per high-power field** on microscopy, with appropriate work-up. **Microscopy of VB3 or EPS to separate IIIA from IIIB is no longer recommended, the distinction having no clinical utility.**
- **Assessment for infection:** midstream urine for culture; if no culture is documented, obtain urinalysis and culture **at the time of symptom recurrence**. **The ICUD recommends the two-glass test using VB2 and VB3.**
 - **A positive midstream (VB2) culture means the diagnosis is category II, not category III.**
 - **The significance of a positive VB3 with a negative VB2 is unclear.** Older classifications considered this diagnostic of CBP, but **asymptomatic men without pelvic pain can have positive localisation cultures.**
 - **In UPOINT, a positive VB3 classifies the patient into the Infection domain and can be an indication for antibiotics.**

- **Atypical organisms:** routine culture not recommended; **nucleic acid amplification testing for gonorrhoea and chlamydia only where there is urethral discharge. Semen cultures are not recommended.**
- **Urine cytology:** optional but indicated where there are irritative voiding symptoms — frequency, dysuria or haematuria. **Carcinoma in situ of the bladder has been diagnosed in patients presenting as CP/CPPS, and haematuria is not always present.**
- **Post-void residual:** check by catheterisation or ultrasound to exclude retention as the cause of symptoms.
- **Blood tests: PSA is not indicated. If measured and elevated, the elevation must not be ascribed to CP/CPPS but evaluated as in any other patient.**

Imaging and Functional Studies

- **Imaging is optional throughout.**
- **CT of the abdomen and pelvis** where there is concomitant abdominal pain, to exclude an intra-abdominal process such as **chronic appendicitis or diverticulitis.**
- **Scrotal ultrasound for testicular pain. TRUS has limited utility.**
- **MRI of the lumbosacral spine** where there are signs and symptoms of lumbar radiculopathy.
- **Routine pelvic MRI is not recommended:** prostatic imaging including gland morphology is **normal in the large majority**, while **extraprostatic abnormalities are found in around two-thirds — most commonly dilatation of the periprostatic venous plexus.** The prostate looks normal; what is found lies outside it and is non-specific.
- **Uroflowmetry:** optional; useful in a **young patient complaining of reduced stream force, as an investigation for stricture disease.**
- **Urodynamics:** optional; consider in patients who fail medical therapy and have significant voiding symptoms, reduced uroflow, or elevated post-void residual.
- **Videourodynamics identifies bladder neck obstruction, in whom intervention may benefit, and diagnoses pseudodyssynergia — retained electrical activity of the external sphincter during voiding in the absence of abdominal straining. The absence of abdominal straining is the definitional element. Brief, intermittent closure of the membranous urethra during voiding may be detected by electromyography or by fluoroscopy. These patients are considered to have pelvic floor dysfunction, so the study routes them to physiotherapy rather than to surgery.**
- **Cystoscopy — two indications: decreased uroflow and/or elevated post-void residual, to evaluate for urethral stricture; and a history of pain with bladder emptying and/or pain relieved by bladder emptying, suggestive of IC/BPS, because a Hunner's lesion may be present and responds to fulguration or intralesional injection.**
- **Prostate biopsy is not recommended for the diagnosis of CP/CPPS alone.**

KEY POINTS: EVALUATION OF CP/CPPS

- CP/CPPS is a diagnosis of exclusion, made only after a thorough search for other causes of pelvic pain.
- CP/CPPS has issues outside the pelvis and outside the usual scope of urology, such as psychological and neurological problems.
- Patients must be assessed for other chronic pain syndromes — IBS, fibromyalgia and chronic fatigue syndrome.
- Patients with tenderness on pelvic examination may benefit from pelvic floor physical therapy and myofascial release.
- Therapy is best delivered multimodally, and UPOINT can direct both evaluation and therapy.

Treatment of Chronic Prostatitis/Chronic Pelvic Pain Syndrome

Antibiotic Treatment

- **Small uncontrolled studies support antibiotics; large randomised placebo-controlled studies have failed to show consistent meaningful benefit.**
- **Levofloxacin 500 mg daily for 6 weeks did not significantly improve NIH-CPSI scores.**
- **The NIH-sponsored trial (Alexander, 2004) randomised patients with an NIH-CPSI score of at least 15 to ciprofloxacin 500 mg twice daily or placebo; scores were not significantly improved at 6 weeks and further change over the subsequent 6 weeks was minimal.**
- **A third trial reported benefit for tetracycline with vitamin C and co-vitamin B, on the premise of activity against nanobacterial infection.**
- **Why antibiotics might work at all**, given that the diagnosis requires UTI to be excluded, has two proposed explanations:
 - **Anti-inflammatory action independent of antibacterial effect** — fluoroquinolones have anti-inflammatory properties, and **ciprofloxacin reduces interleukin-6 and tumour necrosis factor- α production** in cell lines, animal models, and in semen and post-masturbation urine from patients.
 - **Action on bacteria localised to prostatic fluid (VB3 or EPS) without a positive midstream culture** — **category III under the current classification, category II under Drach.** The classification changed; the biology did not.
- **Shorter symptom duration predicts greater antibiotic benefit** — consistent with the central-sensitisation model, in which early disease is still peripherally driven.
- **Meta-analyses disagree by inclusion:** the largest symptom reduction was seen with **antibiotics combined with α -blockers**; a meta-analysis restricted to **the two controlled fluoroquinolone trials found no statistically or clinically significant reduction**; and a **random-effects meta-analysis of all trials concluded that antibiotics are not effective in CP/CPPS.** Restricting to placebo-controlled fluoroquinolone trials gives a negative result; pooling combination therapy gives a positive one.
- **Choice of agent:** fluoroquinolones are the most studied, chosen for spectrum and for pharmacokinetics permitting high prostatic concentration. **Fluoroquinolone-resistant organisms $\rightarrow$ trimethoprim alone or with sulfamethoxazole; second-line agents: tetracycline, doxycycline, macrolides.**
- **EAU recommendation:** antimicrobial therapy (quinolone or tetracycline) for a minimum of 6 weeks, in treatment-naïve patients with CPPS of less than 1 year's duration. Repeated courses of antibiotics in the absence of a positive urine culture are not accepted therapy.

α -Blocker Treatment

- **Rationale is twofold:** some CP/CPPS arises from **dysfunctional voiding**, and **α -adrenergic receptors in the central nervous system are active in long-term pain syndromes** — so the mechanism need not be prostatic at all.
- **Earlier randomised trials were positive** across terazosin, doxazosin, alfuzosin, tamsulosin and silodosin.
- **Two large NIH-sponsored trials were negative:** tamsulosin over 6 weeks, in patients **not necessarily α -blocker naïve**; and alfuzosin over 12 weeks in **α -blocker-naïve patients with shorter symptom duration**, which removes the obvious explanation for the first trial's failure.
- **Meta-analyses either find no benefit, or benefit only in combination** — with antibiotics, or with antibiotics plus anti-inflammatories.
- **ICUD: α -blockers for newly diagnosed, α -blocker-naïve patients with voiding symptoms.** EAU: **α -blockers for CPPS of less than 1 year's duration.** Both converge on early, treatment-naïve disease.

Anti-Inflammatory Therapy

- **Only one-third of symptomatic patients have inflammation on biopsy**, which frames the modest expectations.
- **Rofecoxib** (cyclo-oxygenase-2 inhibitor, withdrawn for cardiac toxicity): **no significant difference from placebo in total or subscale NIH-CPSI; patient global assessment of pain, a secondary outcome, favoured the drug.**
- **Celecoxib in category IIIA: significant reduction in total NIH-CPSI, pain and QOL subscores, but not the urinary subscore** — the pattern of benefit is on pain, not voiding.
- **Zafirlukast added to doxycycline: no difference from placebo.**
- **Prednisolone, a 4-week tapering course from 20 mg to 5 mg daily: no significant change.**
- **Beclometasone dipropionate suppositories** (rectal corticosteroid), observational: **clinically significant improvement in pain and voiding symptoms in the large majority, with no significant adverse effects.**
- **Anti-inflammatory monotherapy is not recommended, but anti-inflammatories may be used within multimodal therapy** — likely ineffective alone, useful combined with an α -blocker and a muscle relaxant having anti-inflammatory properties. **Long-term adverse effects must be weighed.**

5 α -Reductase Inhibitors

- **Finasteride** improved symptoms against placebo in a small pilot using the **Prostatitis Symptom Severity Scale, the precursor to the NIH-CPSI, but not pain**; a 6-month multicentre trial in category IIIA showed improvement that **did not reach statistical significance.**
- **REDUCE** (dutasteride 0.5 mg, 4-year randomised double-blind placebo-controlled prostate cancer risk-reduction trial): **at 48 months, significantly more men on dutasteride achieved a fall of ≥ 6 points in total NIH-CPSI than on placebo, 49% versus 37% — the only positive long-term evidence, and it comes from a trial designed for another purpose.**
- **Pooled analysis shows a significant reduction in NIH-CPSI compared with placebo.**
- **Adverse effects govern selection: reduced ejaculate volume, erectile dysfunction, reduced libido. Best used in older patients with coexisting BPH voiding symptoms** — the patient in whom the drug is indicated anyway.

Medications for Neuropathic Pain

- **Pregabalin: the effect approached but did not reach the primary endpoint; several secondary endpoints significantly improved**, suggesting benefit in a subset.
- **Duloxetine** was beneficial in non-placebo-controlled studies combined with tamsulosin and saw palmetto, or with doxazosin — evidence quality is the limitation.
- **Tricyclic antidepressants are commonly started for suspected neuropathic pain. Amitriptyline is widely used in females with IC, but data in males are lacking. In IC/BPS studies, better response required escalation to at least 50 mg** — underdosing is the common reason for apparent failure.
- **Tanezumab** (monoclonal antibody against nerve growth factor), phase IIa: **marginal improvement in average daily pain and urgency frequency at 6 weeks, with no difference in overall NIH-CPSI.**
- **A fatty acid amide hydrolase inhibitor** — the enzyme that breaks down endogenous cannabinoids — **gave no improvement in pain, though voiding frequency improved across all groups.**

Conservative Measures

- **Roughly half of patients report sensitivity to specific foods, most commonly spicy foods, coffee, tea, chilli and alcoholic beverages. Items reported to improve symptoms — docusate, psyllium, water, herbal teas and polycarbophil — are all bowel-directed, which fits the IBS overlap. There are no dietary recommendations applicable to all patients; diet must be individualised.**

- **Aerobic exercise produced significant improvement in pain, QOL and total NIH-CPSI compared with stretching over 18 weeks.**
- **Stress reduction alone produced improvement in the large majority in an early series.** Options include conventional therapy with a psychologist or psychiatrist; **an 8-week cognitive therapy programme significantly reduced pain, disability and catastrophising**, and **internet delivery is a promising route to wider availability.**

Acupuncture

- **Mechanism unknown; postulated to ameliorate neuropathic pain. Typically four to six needle points.**
- **Superior to sham needling placed away from true acupuncture sites, with increases in β -endorphin and leucine enkephalin offering a biochemical correlate. Adding electrical stimulation is superior to sham and to advice plus exercise alone.**
- **A large recent trial showed a substantially higher responder proportion than sham at week 32 — 61.5% versus 38.3%, adjusted odds ratio 2.6 — the strongest evidence for any conservative measure in this chapter. A meta-analysis found acupuncture superior to both sham and Western medicine across pain score, NIH-CPSI, QOL, urinary symptoms and efficacy rate.**
- **Given few side effects, acupuncture can be offered to those who fail first-line medical therapy or who prefer not to take medication.**

Pelvic Floor Physical Therapy and Skeletal Muscle Relaxants

- **Biofeedback and pelvic floor retraining — learning to contract and then relax the pelvic muscles selectively — have shown significant improvement in small series.**
- **Pretreatment urodynamics and sphincter electromyography do not predict outcome — do not use them to select patients.**
- **Biofeedback and pelvic floor therapy are effective in patients with pseudodyssynergia on urodynamics.**
- **The Stanford protocol — myofascial trigger point release with paradoxical relaxation training (essentially biofeedback) — has reproducibly shown benefit in uncontrolled studies, including improvement in sexual function: ejaculatory pain, reduced libido and ED.**
- **A condensed 6-day course of myofascial release and trigger point manipulation produced a fall of ≥ 6 points in total NIH-CPSI at 6 months in 60% of those completing it.**
- **The methodological difficulty is the control: a collaborative network study established global massage as the control for myofascial physical therapy, and although not powered as a definitive outcome study, myofascial therapy gave a significantly greater global response than therapeutic massage, 57% versus 21%.**
- **Refer to a physical therapist experienced in the techniques where possible, since outcomes are better at specialised centres.**

Surgical Therapy

- **Transurethral resection is not expected to improve pain, because of centralisation of pain and the absence of evidence that the prostate is a primary pain source.**
- **The one entity that matters in a subset is bladder neck hypertrophy. There are no pathognomonic diagnostic criteria; suspect it where chronic pelvic pain coexists with significant obstructive LUTS. Pressure-flow urodynamics show raised voiding pressure with reduced flow and relative narrowing of the bladder neck.**
- **Pseudodyssynergia must be excluded first — it is treated by physiotherapy and biofeedback, not by incision. Bladder neck incision produced significant improvement in almost all patients in a small uncontrolled series.**

- **Neurostimulation: sacral nerve stimulation (SNS) and percutaneous tibial nerve stimulation (PTNS) are approved for urinary symptoms but not specifically for pelvic pain** — their use in pain is off-label. Most SNS evidence is case series in IC/BPS; **implantation follows test stimulation showing >50% symptom reduction**, with implantation rates of 56–68% and durable significant improvement in 64–72% at up to 5 years. PTNS versus sham in CP/CPPS gave significant reduction in pain, urgency and NIH-CPSI at 12 weeks. Transcutaneous electrical nerve stimulation succeeded in about half at 12 weeks, sustained over a mean follow-up of about 3.5 years. **Offer PTNS for CPPS, and SNS where pelvic pain coexists with frequency and urgency.**
- **Electromagnetic stimulation:** seated on an electromagnetic chair for **30 minutes twice weekly for 6 weeks**; a majority achieved a fall of more than 6 points in NIH-CPSI at 24 weeks, with improvement in total and pain subscales and in the International Prostate Symptom Score, and an earlier study showed benefit over sham at 3 months and 1 year.
- **Cystoscopy and fulguration of Hunner's lesions** for patients whose pain is of bladder origin; **although not commonly found, fulguration provides symptom relief.**
- **Not recommended — radical prostatectomy:** surgical removal of painful structures where pain exists elsewhere in the pelvis is not recommended, and radical prostatectomy is specifically not recommended in the absence of treatment for prostate carcinoma — the same reasoning as for transurethral resection.

Cochrane Review of Non-Pharmacological Interventions

- Examined **acupuncture, circumcision, the electromagnetic chair, lifestyle modification, physical activity, prostatic massage, extracorporeal shockwave lithotripsy and transrectal thermotherapy**; inadequate information was found on psychological therapy and on prostatic surgery.
- **Conclusion:** some interventions decrease symptoms in an appreciable number of patients without increased adverse events, but the quality of evidence was mostly low. The reviewers called for better-reported trials.

Treatment: Summary and Approach

- **The governing limitation:** there are no positive clinical trials for monotherapy in CP/CPPS. Every failed trial in this chapter tested a single agent across an unphenotyped population.
- **Start with the most conservative treatments possible, including lifestyle change.**
- **Then deliver simultaneous multimodal therapy directed by the patient's individual phenotype** — not sequential single-agent trials.
- **Myofascial pain and the need to address the pelvic floor are emerging as among the most important aspects of treatment and should be high on the differential.**
- **Most patients need more than one specialist:** urology plus neurology, gastroenterology, psychiatry or psychology, and physical medicine and rehabilitation.

KEY POINTS: TREATMENT OF CP/CPPS

- Repeated courses of antibiotics without a positive urine culture are not indicated.
- The best treatment combination from meta-analyses is antibiotics plus α -blockers.
- Patients with bladder pain should be considered for bladder-specific therapies.
- Patients with tenderness on pelvic floor examination should be referred for pelvic floor physical therapy.
- Sacral neuromodulation can be used in patients with pain who also have frequency and urgency.

- Treatment is best undertaken as simultaneous multimodal therapy addressing all aspects of the patient's symptoms.

ACUTE ORCHITIS AND EPIDIDYMITIS

Acute Orchitis

- **The incidence of acute orchitis is not known.** In a review of acute scrotum presentations, the distribution was epididymo-orchitis 28.7%, isolated epididymitis 28.4% and orchitis 10.3% — isolated orchitis is the least common of the three. Mean age in the orchitis and epididymo-orchitis groups was in the mid-forties.
- **Nomenclature:** *orchidoepididymitis* and *orchiepididymitis* have both been used; **the chapter uses epididymo-orchitis.**

Aetiology

- Acute orchitis divides into specific orchitis and viral orchitis.
- **Specific orchitis** is caused by a wide range of organisms, including by the haematogenous route. Over the age of 35, orchitis and epididymo-orchitis are most commonly bacterial and predominantly Gram-negative — *E. coli* and *Pseudomonas* — mirroring the age-and-organism distribution of acute epididymitis.
- **Other agents:** *Treponema pallidum*, *M. tuberculosis*, and organisms uncommon in Western practice including *Brucella* and filariasis. Candiduria may also lead to epididymo-orchitis.

Viral orchitis.

- **Mumps virus is the commonest viral cause.** Vaccination had almost eliminated the disease by 2001, but sporadic outbreaks continue and involve a high proportion of previously vaccinated persons — vaccination history does not exclude mumps.
- **Clinical sequence:** prodrome of headache, malaise and myalgia → parotitis developing 2–3 weeks after exposure → orchitis, usually with epididymitis and fever, resolving within 1 week.
- **Testicular atrophy** follows in half of cases, and may be associated with oligospermia and reduced fertility, but usually not complete infertility.
- **Other viruses:** HIV-1 — by direct testicular infection and as a consequence of immunosuppression; Epstein-Barr virus — orchitis is a feature of infectious mononucleosis, indicating a direct testicular effect; herpes simplex virus type 2, detected in the testis; and dengue, smallpox and chickenpox.
- **SARS-CoV-2** has been linked to testicular pain, orchitis and epididymitis. **Proposed mechanism:** viral binding to the angiotensin-converting enzyme 2 receptor, expressed on seminiferous tubule cells and on Sertoli and Leydig cells, producing tissue inflammation. **Current evidence is mostly case reports describing relatively benign courses managed conservatively, irrespective of pulmonary manifestations,** though one report described **necrotising epididymo-orchitis requiring orchidectomy** with virus confirmed in the tissue. **Vaccination was associated with significantly reduced odds of developing orchitis or epididymitis.** Implications for spermatogenesis and fertility remain unresolved.

Autoimmune orchitis.

- **Definition:** autoimmune aggression against the testis characterised by anti-sperm antibodies (ASAs).
- **Primary:** ASAs present, no associated systemic disease, asymptomatic presentation — encountered principally in the evaluation of male infertility.
- **Secondary:** symptomatic orchitis associated with systemic disease — usually a primary vasculitis, particularly Behçet disease, polyarteritis nodosa and Henoch-Schönlein purpura.

- **Mechanism:** although the testis is an **immune-privileged site**, the **blood–testis barrier is altered by a T cell-mediated inflammatory response** with increased **interferon- γ , interleukin-1, -6, -12, -17 and -23**, and consequent ASA production.

Evaluation

- **History:** circumstances of onset, sexual history, recent instrumentation or surgery, prior episodes, and trauma.
- **Duration of pain is critical, because testicular torsion is a prime differential diagnosis.**
- **Ask specifically about upper respiratory symptoms and parotitis suggesting mumps**, and about **fever, sweats, headache and back pain suggesting brucellosis**; and about **the conditions causing secondary autoimmune orchitis**.
- **Examination:** signs of systemic involvement — fever, tachycardia; **the scrotum usually shows a tender testis and spermatic cord**.
- **Urine culture in all patients** — positive in a little under 40% of both isolated orchitis and epididymo-orchitis. **Test sexually active patients, especially those with epididymal involvement, for STIs.**
- **Scrotal ultrasound distinguishes testicular torsion from orchitis and identifies testicular tumours and abscesses.**

Treatment

- **Signs of infectious orchitis:** start antibiotics against Gram-negative uropathogens and adjust to urine culture.
- **Suspected STI, especially younger patients with concomitant epididymitis:** treat accordingly.
- **Mumps orchitis** can be treated with interferon, but this does not always prevent testicular atrophy.
- **Secondary autoimmune orchitis:** corticosteroids, immunosuppressants such as azathioprine or cyclophosphamide, and intravenous immunoglobulin.
- **HIV:** initial antibiotic treatment followed by maintenance suppression, particularly where *Salmonella* is the organism.
- **Follow-up is mandatory, especially where symptoms do not resolve, to exclude missed torsion or testicular tumour.**
- **Counsel that acute pain may resolve in 1–3 days but the ensuing inflammation can take 2–4 weeks to settle completely** — failure to give this advice generates unnecessary re-presentations.

Acute Epididymitis

- **Acute epididymitis:** pain, swelling and inflammation of the epididymis, of less than 6 weeks' duration.
- **Common** — incidence 25 to 65 per 10,000 person-years, accounting for up to 600,000 outpatient cases annually in the United States. Age is variable; it affects both paediatric and adult patients.

Aetiology

- **The accepted route is ascent of organisms from the urethra**, first described by Campbell in 1927 in gonococcal urethritis with subsequent epididymitis, and **confirmed by approximately 80% concordance between simultaneous urine or urethral and epididymal isolates**.
- **Causative organisms depend on age** — the classic teaching: over 35 years and in children, the organisms are those causing UTI, e.g. *E. coli*; under 35 and sexually active, *Chlamydia trachomatis* and *Neisseria gonorrhoeae*.
- **Risk factors in older patients without sexually transmitted epididymitis:** prior prostatic infection or UTI, recent urinary tract surgery, **bladder outlet obstruction from BPH or stricture**, uncircumcised penis or congenital urinary tract deformity, and immunosuppression.

- **Contemporary microbiology with updated diagnostics:** a bacterial pathogen was identified in **87%**, with *E. coli* the commonest; the commonest sexually transmitted organism was *Chlamydia*, followed by *Mycoplasma* and *N. gonorrhoeae*. Four conclusions follow:
 - The age cut-offs are not strict rules — over 40% of those under 35 grew enteric organisms.
 - All sexually active individuals should be screened for STI, since the sexual history does not reliably match the microbiology.
 - Viruses were not a significant contributing factor, and mumps was not detected.
 - 13% had no identifiable microbial cause.

Clinical Evaluation

- Pain and swelling begin at the cauda of the epididymis and ascend to involve the rest of the epididymis and the testis — the characteristic pattern.
- Presentation is unilateral in 96% of cases. Fever above 38 °C in about a quarter; epididymal abscess in under 10%; dysuria in one-third.
- Examination: systemic signs — fever, tachycardia; a tender epididymis and spermatic cord; induration of the cauda or whole epididymis is frequent; orchitis is present in half of cases.
- Investigations: full blood count; urine for routine culture and susceptibility; assessment of inflammation by leucocyte esterase on first-void urine or microscopy of spun first-void urine sediment.
- In older patients, measure post-void residual to exclude retention as the cause of the UTI and consequent epididymitis — treat the cause, not just the epididymis.
- Differential diagnosis of the acute scrotum: testicular torsion, torsion of the appendix testis, inguinal hernia, testicular carcinoma, varicocele pain, scrotal or testicular abscess, acute orchitis, testicular trauma.
- Scrotal ultrasound may show hydrocele, epididymal enlargement, hyperperfusion and testicular involvement; its critical role is confirming the diagnosis and excluding torsion.

Treatment and Follow-Up

- Symptomatic measures: bed rest, scrotal elevation, local cooling, and NSAIDs.
- Most patients are managed as outpatients; admit those with significant leucocytosis and fever.
- Antimicrobial therapy by likely pathogen:
 - Chlamydia or gonorrhoea likely: ceftriaxone 500 mg intramuscularly as a single dose plus doxycycline 100 mg orally twice daily for 10 days (ceftriaxone 1 g if body weight ≥ 150 kg).
 - Chlamydia, gonorrhoea or enteric organisms likely — those practising insertive anal intercourse: ceftriaxone 500 mg intramuscularly single dose plus levofloxacin 500 mg orally daily for 10 days.
 - Enteric organisms only: levofloxacin 500 mg orally daily for 10 days.
- Confirmed chlamydia or gonorrhoea: test for other STIs including HIV, and refer sexual partners for evaluation and presumptive treatment.
- Re-evaluate if symptoms do not resolve within 72 hours. Repeat ultrasound in non-responders to exclude progression to abscess or testicular infarction, either of which requires surgery.
- Follow urine culture susceptibilities closely, given rising fluoroquinolone resistance.

Chronic Orchitis and Epididymitis

Chronic Orchitis

- **Chronic orchitis** is not well defined. In the absence of known chronic infection or inflammation, **chronic orchialgia** or **chronic scrotal content pain** is the more appropriate classification for testicular pain or discomfort exceeding 3 months.
- **Granulomatous orchitis** — granulomatous inflammation of the testis, with causes including **tuberculosis**, **brucellosis**, **actinomycosis**, **syphilis**, **leprosy** and **sarcoidosis**.
- **Idiopathic granulomatous orchitis** — rare, of unknown cause, with non-specific granulomatous inflammation, multinucleated giant cells and no caseation. The diagnosis is usually made after orchidectomy for suspected malignancy, because the ultrasound appearance is similar. Presentation may be chronic pain or painless testicular enlargement.
- **IgG4-related disease** affects testicular cells and perivascular tissue, causing fibrosis, and **IgG4-positive cells** have been found in idiopathic granulomatous orchitis specimens in patients with no systemic IgG4 disease.
- **Malakoplakia** — distinguished from granulomatous orchitis by a dense histiocytic infiltrate with pathognomonic Michaelis–Gutmann bodies.

Chronic Epididymitis

- **Definition:** discomfort and/or pain of at least 3 months' duration localised to one or both epididymides — the same threshold as orchialgia.
- **Uncommon in urological practice** — around 1% of patients seen, with a mean symptom duration of 2.5 years and mean age around 41.
- **Nickel classification** — three categories: **inflammatory**, **obstructive**, and **epididymalgia (no clear cause)**.
 - **Inflammatory:** infective (chlamydia), post-infective (after acute bacterial epididymitis), granulomatous (tuberculosis), drug-induced (amiodarone), and idiopathic inflammatory.
 - **Obstructive:** congenital, acquired or iatrogenic obstruction — post-infective or post-surgical, **including post-vasectomy**.
- **Patients with AIDS** are prone to tuberculous epididymitis.

Tubercular Epididymitis

- **Medical treatment:** four antibiotics simultaneously for a minimum of 6 months; five different chemotherapy regimens exist depending on the form of urogenital tuberculosis.
- **Surgery** may be needed to drain an abscess or to remove a chronically infected epididymis or testis. All surgery must be performed while on anti-tuberculous medication, and the subsequent treatment duration is determined by the histology of the removed tissue.
- **Indication for surgery:** failure of a scrotal mass to respond, or an increase in size, after 3 weeks of anti-tuberculous chemotherapy. Such a mass must be explored through an inguinal incision to exclude testicular or epididymal malignancy.
- **Presentation** is subacute, with epididymal swelling that may be painless; systemic signs of tuberculosis, scrotal thickening and a characteristic fistula may be present.
- **Tuberculosis** characteristically involves the epididymis, which is the sole site in 45% of cases, with testicular involvement usually by contiguous spread.

Evaluation and Treatment of Chronic Epididymitis

- **History** must include any treatment for acute epididymitis. A Chronic Epididymitis Symptom Index has been published.

- **Examination findings map onto the Nickel categories** — the practical value of the classification: **inflammatory** — abnormal swelling or induration; **obstructive** — may have induration; **epididymalgia** — the epididymis is palpably normal.
- **Urinalysis and midstream urine culture**; evaluate for STI if there is urethral discharge; **scrotal ultrasound** to look for epididymal and testicular tumours; **assess for neurological contributors**, especially back pain indicating disc disease.
- **Conservative measures first**. Excluding malignancy permits counselling on the benign nature of the condition and observation where symptoms are mild. **Scrotal support and warm compresses**.
- **Medical treatment lacks evidence-based studies**; agents used include antibiotics, anti-inflammatories, phytotherapy and anxiolytics. **About a quarter of patients receive analgesics**.

Chronic Scrotal Content Pain

- **Chronic scrotal content pain (CSCP)**: unilateral or bilateral pain localised to structures within the scrotum for at least 3 months that interferes with daily life.
- It encompasses chronic orchitis/orchialgia/testalgia and chronic epididymitis/epididymalgia. There are no standardised, evidence-based guidelines for its diagnosis or management.

Epidemiology

- True prevalence is difficult to estimate, given the multiplicity of causes and the large idiopathic proportion.
- Orchialgia accounts for 2.5–4.8% of European clinic visits; approximately 100,000 cases per year are estimated in the United States; almost 40% of chronic pelvic pain localises to the scrotum.
- Commonest age at presentation: 20 to 30 years.
- Post-procedural causes by frequency: inguinal hernia repair 18%, vasectomy 6–12%, scrotal surgery up to 5%, abdominal or groin surgery 1–2%.

Aetiology and Pathophysiology

- **Vasectomy**: short-term scrotal pain lasting weeks in about 30%; long-term pain prompting treatment in approximately 1 in 1000 — the figure for consent.
- **Inguinal hernia repair**: iatrogenic injury in up to 14%, with greater risk after surgery for recurrent hernia or where mesh was used.
- **Other causes**: varicocele or varicocele surgery, referred pain from a mid or distal ureteric stone, indirect hernia, aortic or iliac aneurysm, trauma, perineural fibrosis.
- **Pelvic floor dysfunction is a major and under-recognised contributor**: 93% of chronic orchialgia patients had at least one symptom of pelvic floor dysfunction and 88% had increased resting pelvic floor muscle tone — the same mechanism as in CP/CPPS.
- **Approximately 18–25% of chronic orchialgia is idiopathic**.
- **The underlying pathophysiology is neuropathic**, by two processes:
 - **Central sensitisation** — increased neuronal activity in central nociceptive pathways producing pain hypersensitivity.
 - **Wallerian degeneration** — found on nerve biopsy in 84% of patients with pain versus 20% of controls, in reproducible patterns involving the cremasteric muscle layer, perivascular tissues, and posterior periaxillary/lipomatous tissues — the "trifecta nerve complex". It is a coordinated response to nerve injury — Schwann cell proliferation, extracellular matrix change, and recruitment of neurotrophins and cytokines to facilitate axonal regeneration — and the ensuing neuroinflammation can produce

neuropathic pain. Because a fifth of asymptomatic controls also showed degeneration, it is not invariably symptomatic.

- **Calcitonin gene-related peptide, a marker of sensory nociceptors, is increased in the spermatic cords of patients with pain.**

Evaluation

- **The first determination: is the pain confined to the testis and scrotum, or part of a wider pelvic pain problem consistent with CP/CPPS? This decides the entire management pathway. The diagnosis is otherwise directed at excluding the other causes of chronic orchialgia.**
- **History:** intermittent torsion; prior testicular or epididymal infection; **bowel dysfunction with constipation or diarrhoea suggesting IBS; low back pain and vertebral disc disease, since orchialgia can be referred from the back;** LUTS indicating voiding dysfunction predisposing to UTI.
- **Screening for low testosterone or vitamin B12 found the majority of chronic orchialgia patients deficient in one or both — an inexpensive and often-omitted test pair.**
- **The Chronic Orchialgia Symptom Index (COSI): a validated 12-item instrument covering pain, sexual symptoms and quality of life, derived from an original 70-item questionnaire, developed to standardise assessment and treatment response in CSCP.**
- **Examination: thorough examination of the genitalia including testes, epididymis, prostate and pelvic floor is mandatory; exclude inguinal hernia.**
- **Investigations: urinalysis; scrotal duplex Doppler ultrasound to assess for anatomical or congenital abnormality or a mass; CT if a ureteric stone is suspected.**
- **Spermatic cord block (SCB) is supported as part of the initial assessment. Technique: 20 mL of 0.25% bupivacaine injected into the spermatic cord at the level of the pubic tubercle. Purpose: determine whether nerves within or surrounding the cord are part of the pain process, and predict the effectiveness of denervation. A temporary pain reduction exceeding 50% is an independent predictor of response to microsurgical denervation of the spermatic cord (MDSC).**

Pharmacologic and Minimally Invasive Treatment

- **Antibiotics are commonly used but rarely indicated, exactly as in CP/CPPS. Doxycycline and quinolones are the agents recommended when used, for their penetration into scrotal structures and because they can be given for up to 4 weeks.**
- **Anti-inflammatories are used for analgesia, with little evidence for a specific therapeutic target.**
- **Neuropathic agents follow from the neuropathic pathophysiology: tricyclic antidepressants — amitriptyline and nortriptyline — and the anticonvulsants gabapentin and pregabalin. Chronic opioid use should be discouraged.**
- **Therapeutic SCB: beyond its diagnostic role the block can be therapeutic for those unable or unwilling to proceed to MDSC, though the effect may be temporary and require repeat injection. A series of four additional blocks with bupivacaine and triamcinolone after a positive diagnostic block gave sustained pain relief in about 70% and complete response in about 20% at a mean follow-up of 16 months.**
- **Pelvic floor physical therapy should be considered in patients with orchialgia and abnormal tone on DRE — the DRE finding is the selection criterion. Twelve sessions gave pain improvement in half and complete resolution in a minority.**
- **OnabotulinumtoxinA: cord injection 1–2 cm distal to the external ring, targeting branches of the genitofemoral and ilioinguinal nerves, in patients who had already had temporary relief from an SCB. Open-label results were encouraging at 1 and 3 months, but most patients had return of pain by 6 months, and a double-blind randomised controlled trial found no difference between local anaesthetic**

alone and local anaesthetic plus onabotulinumtoxinA — at double the open-label dose. **The open-label benefit is therefore attributable to placebo effect**, the clearest demonstration in this chapter of why uncontrolled scrotal pain series must be read with caution.

- **Pulsed radiofrequency (PRF) of the spermatic cord**, borrowed from chronic back pain and visceral pelvic pain: **3 minutes at 42 °C** in patients who responded to a local anaesthetic block. A small series showed **marked reduction in pain scores with no testicular atrophy or loss of scrotal or penile skin sensation**. **PRF to the ilioinguinal nerve and the genital branch of the genitofemoral nerve in chronic post-surgical orchialgia** gave more than 50% pain reduction in 80% versus 20% in a sham group. **No long-term results are available**.
- **Additional options: vibratory stimulation** reduced average and maximum pain scores and pain frequency in a small proof-of-concept study; **transcutaneous electrical nerve stimulation plus analgesia** gave significant reduction in pain and improvement in QOL over analgesia alone after 4 weeks.

Surgical Treatment

Neuromodulation.

- **Sacral nerve stimulation via the caudal epidural space produced substantial pain reduction in a patient with recurrent epididymitis**. The anatomical justification for a testicular effect from an S2–S4 device is that **dermatomes are larger than most textbooks describe** and **around a quarter of pudendal nerves receive a contribution from S1**.
- **Spinal cord stimulation in the epidural space at T10–T11** has been reported to help, as has **cutaneous stimulation of the cutaneous branch of the ilioinguinal and the genital branch of the genitofemoral nerves**.

Surgery for identifiable intrascrotal lesions.

- **Where pain localises to a specific lesion, directed surgery has high success: hydrocelectomy 100%, spermatocelectomy 94%, and orchidopexy for intermittent torsion 89%.**
- **The commonest target-directed operation is spermatic vein ligation for a painful varicocele: complete resolution 85% with primary microsurgical repair versus 72% non-microsurgical, 81% for laparoscopic repair with a 6.8% recurrence rate, and 90% for microsurgical salvage of recurrent or persistent varicocele. Percutaneous embolisation can be effective but recurrence rates are generally higher than after microsurgical repair.**
- **Predictors of response to varicocelectomy: more veins ligated (>7), higher preoperative pain score (>6), and longer duration of pain (>9 months).**

Vasovasostomy or vasoepididymostomy for post-vasectomy pain.

- **An option unique to post-vasectomy pain syndrome — reverse the vasectomy and restore vasal patency.**
- **Complete response 34–50%; overall improvement 82–93%; reoperation rate about 20% over 1–9 years.**
- **Pain relief may take 3 months or more — essential for counselling and expectation management. Given relatively minimal morbidity, it is attractive for persistent long-term post-vasectomy pain with no other identifiable cause.**

Microsurgical denervation of the spermatic cord.

- **First described by Devine and Schellhammer in 1978.**
- **Principle: transect the nerves of the spermatic cord while preserving the blood supply and the lymphatics, the latter to prevent hydrocele formation. The internal spermatic veins are ligated; the cremaster muscle and spermatic cord fascia are preserved.**

- **The trifecta nerve complex — the three sites of highest Wallerian degeneration — is the cremasteric muscle fibres, the perivasal sheath and the posterior lipomatous tissue, forming the anatomical basis for relief from denervation.**
- **The targeted "trifecta" MDSC ligates only the cremasteric fibres, the sheath of the vas deferens, and the posterior cord perivascular adipose tissue. Compared with full MDSC it gives no difference in pain improvement or failure rates, with operative time reduced from about 53 to about 21 minutes.**
- **A robotic approach has been described, including a biowrap around the cord to minimise scarring and neuroma formation.**
- **Reported cure rates across series: 50–70%. The largest series achieved complete resolution in 49% and more than 50% pain reduction in a further 34% at a median follow-up of 24 months, with improvement continuing to progress over time — higher rates at 4 years than at 6 months.**
- **Involvement of multiple scrotal structures — testis, epididymis and spermatic cord — predicted higher success (81%). Prior failed surgery for scrotal pain does not compromise the result.**
- **Patient selection: a positive SCB predicts success, and the degree of relief after the block correlates with relief after surgery. Consider MDSC in patients with at least 50% pain reduction after 20 mL of 0.25% bupivacaine at the pubic tubercle. Some patients obtain more relief from surgery than from the block itself.**
- **Complications are few, but testicular atrophy is among them and must be discussed.**

Epididymectomy.

- **Results vary widely across series, but most report complete improvement in 50–75% and further improvement in 15–21%.**
- **Better results where there is a palpable area of tenderness, an ultrasound abnormality, or a cyst — a localised target predicts success.**
- **Results specifically for post-vasectomy pain are mixed, with better rates where the distal portion of the vasectomy site is included in the specimen, since this removes perineural inflammation, sperm extravasation and fibrosis.**
- **Progression to orchidectomy for persistent pain after epididymectomy ranges widely across series.**

Orchidectomy.

- **Pain relief is reported in 0–75%, with partial improvement in 20–43% — the widest and least reliable range of any procedure here.**
- **The inguinal approach gives a higher complete response than the scrotal approach, 75% versus 55%.**
- **The critical predictor: patients operated on for nociceptive pain from testicular injury such as ischaemia respond better than those operated on for neuropathic pain from testicular nerve injury.**
- **Orchidectomy for persistent pain after MDSC gave complete response in 70%, with 30% still requiring analgesia. Bilateral scrotal pain was a risk factor for pain after MDSC.**
- **Even as a last resort, "phantom" pain may persist.**
- **Management of pain persisting after orchidectomy mirrors that for persistent pain after inguinal hernia repair: identify the genital branch of the genitofemoral nerve at the external ring, resect the neuroma, and place the proximal nerve end within the pelvis — with excellent relief reported in most patients.**

KEY POINTS: ORCHITIS, EPIDIDYMITIS AND SCROTAL PAIN

- **Viral orchitis is most commonly caused by the mumps virus.**
- **Acute epididymitis is more commonly caused by a sexually transmitted infection under 35 and by enteric bacteria over 35.**

- For scrotal pain, a duplex Doppler ultrasound should be performed to assess for anatomical or congenital abnormality, torsion, or mass.
- A diagnostic spermatic cord block should be considered as part of the assessment of chronic scrotal content pain; it can also be therapeutic, and a positive response predicts success of microsurgical denervation.
- Pelvic floor physical therapy should be considered in patients with orchialgia and abnormal tone on digital rectal examination.
- The cremasteric muscle fibres, the perivasal sheath and the posterior lipomatous tissue are the three areas to be addressed during microsurgical denervation of the spermatic cord.